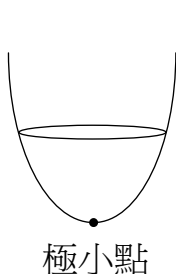
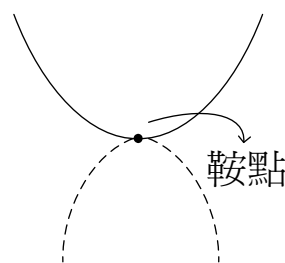
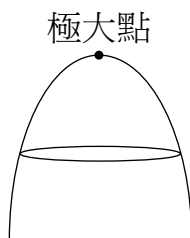


§14.7 Maximum and Minimum Values

Homework : 5,11,17,29,33,37,41,45,51,53,54



(Minimum in all directions)



(saddle point)

(One direction is a maximum and another direction is a minimum.)

(I) How to find such points?

(1) Find critical points : (a, b)

(a) $f_x(a, b) = f_y(a, b) = 0$. or

(b) f_x or f_y does not exist at (a, b) .

(2) 2nd Derivative Tests.

Let $\alpha = f_{xx}(a, b)$, $\beta = f_{xy}(a, b) = f_{yx}(a, b)$, $\gamma = f_{yy}(a, b)$

Set $D = \begin{vmatrix} \alpha & \beta \\ \beta & \gamma \end{vmatrix} = \alpha\gamma - \beta^2$

(i) $D > 0$, $d > 0$ (or $\gamma > 0$) ($\because D > 0 \Rightarrow \alpha, \gamma$ 同號).

$\Rightarrow f(a, b)$ local minimum and (a, b) is a minimum point.

(ii) $D > 0$, $\alpha < 0$ (or $\gamma < 0$) $\Rightarrow f(a, b)$ local max.

(iii) $D < 0$: (a, b) is a saddle point.

(iv) For any other cases, the test fails.

Why : ?

(2-i) 的條件中

$\alpha > 0 \Rightarrow$ 在 x 軸方向 $(a, b, f(a, b))$ 是凹向上.

$D > 0 \Rightarrow$ 其它所有方向也凹向上.

Proof :

$$u = \langle h, k \rangle .$$

$$D_u f(a, b) = f_x h + f_y k .$$

$$\begin{aligned} D_u^2 f &= D_u (D_u f) = (f_{xx} h + f_{xy} k) h + (f_{yx} h + f_{yy} k) k \\ &= f_{xx} h^2 + 2f_{xy} hk + f_{yy} k^2 \\ &= \alpha h^2 + 2\beta hk + \gamma k^2 \\ &= \alpha \left(h + \frac{\beta k}{\alpha} \right)^2 + \frac{k^2}{\alpha} (\alpha\gamma - \beta^2) . \end{aligned}$$

If $\alpha > 0 + D > 0 \Rightarrow D_u^2 f > 0$ for any direction u .

$\Rightarrow$ All directions at $(a, b, f(a, b))$ are concave up.

Example 1 : $f(x, y) = x^4 + y^4 - 4xy + 1$. Find local max, min and saddle point.

Solution :

$$\begin{cases} f_x = 4x^3 - 4y = 0 \\ f_y = 4y^3 - 4x = 0 \end{cases}$$

$\Rightarrow (0, 0), (1, 1), (-1, -1)$ are critical points.

$$f_{xx} = 12x^2 \quad f_{xy} = -4 \quad f_{yy} = 12y^2$$

$$D(x, y) = 144x^2y^2 - 16$$

$$\Rightarrow D(0, 0) = -16 < 0.$$

$\Rightarrow (0, 0)$ is a saddle point.

$$D(1, 1) = 128 > 0, \quad f_{xx}(1, 1) = 12 > 0$$

$\Rightarrow f(1, 1) = -1$ is a local min.

$$D(-1, -1) = 128 > 0, \quad f_{xx}(-1, -1) = 12 > 0$$

$$\Rightarrow f(-1, -1) = -1 \text{ is also a local min.}$$

(II) Theorem : f is continuous on a closed and bounded set D .

$\Rightarrow$ (i) There exist absolute minimum and absolute maximum.

(ii) Those extreme points occur at the critical points of f or the boundary of D .

Example 2 : $f(x, y) = x^2 - 2xy + 2y$ on $0 \leq x \leq 3, 0 \leq y \leq 2$.

Find absolute max and min.

Solution :

$$\begin{aligned} f_x &= 2x - 2y \\ f_y &= -2x + 2 \end{aligned} \Rightarrow x = 1, y = 1.$$

(x, y)	$(1, 1)$	$x = 0$ $0 \leq y \leq 2$	$x = 3$ $0 \leq y \leq 2$	$y = 0$ $0 \leq x \leq 3$	$y = 2$ $0 \leq x \leq 3$
$f(x, y)$		$2y$	$9 - 4y$	x^2	$(x - 2)^2$
values	1	0,4	1,9	0,9	0,4

$\Rightarrow 0$: absolute minimum.

9 : absolute maximum.

Example 3 : $f(x, y) = 2x^2 + x + y^2 - 2, x^2 + y^2 = 4$. Find absolute extreme.

Solution :

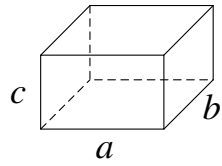
$$\begin{aligned} f_x &= 4x + 1 \\ f_y &= 2y \end{aligned} \Rightarrow x = -\frac{1}{4}, y = 0.$$

(x, y)	$\left(-\frac{1}{4}, 0\right)$	$x^2 + y^2 = 4$	
$f(x, y)$		$x^2 + x - 2$ $-2 \leq x \leq 2$	
values	$-\frac{17}{8}$	$0, 4, -\frac{9}{4}$	

$\Rightarrow \max = 4$ occurs at $x = 2, y = 0$.

$\min = -\frac{9}{4}$ occurs at $x = -\frac{1}{2}, y = \pm \frac{\sqrt{15}}{2}$.

Example 4 : A box with open top. with constraint :



$$ab + 2bc + 2ac = 12$$

Find max volume.

Solution :

$$\frac{ab + 2bc + 2ac}{3} \geq \sqrt[3]{4a^2b^2c^2}$$

$$\Rightarrow \quad 64 \geq 4a^2b^2c^2$$

$$\Rightarrow \quad 4 \geq abc$$

Maximum volume = 4 when $ab = 2bc = 2ac$

$$\Rightarrow a = 2, b = 2, c = 1.$$

Example 5 : (x, y, z) satisfies $x + 2y + 3z = 6$.

Find max $V = xyz$.

Solution :

$$\frac{6}{3} \geq \sqrt[3]{6xyz} \Rightarrow xyz \leq \frac{4}{3}$$

$$\text{Max } V = \frac{4}{3} \text{ when } x = 2y = 3z \Rightarrow x = 2, y = 1, z = \frac{2}{3}.$$